Single-electron transistor

Part A. Tunneling effect (3.0 points)

A.1 (0.1pt)

$W =$ $\Delta W =$

A.2 (0.3pt)

$q_g =$ $q_j =$

A.3 (0.3pt)

$\Delta E =$

A.4 (0.5pt)

$A =$ $Q =$

A.5 (0.5pt)

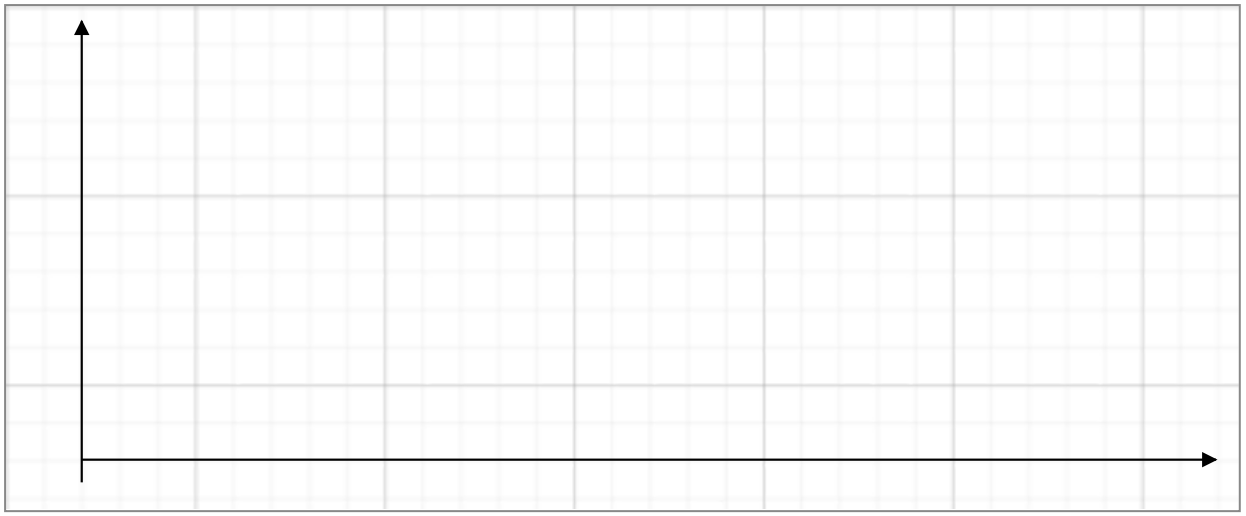
$I_t =$

Condition on n:

A.6 (0.3pt)

$I =$

A.7 (1.0pt)
Plot:



Part B. Single-electron transistor (4.5 points)

B.1 (0.2pt)

$\varphi =$

B.2 (1.0pt)

$A =$

B.3 (1.0pt)

$Q =$

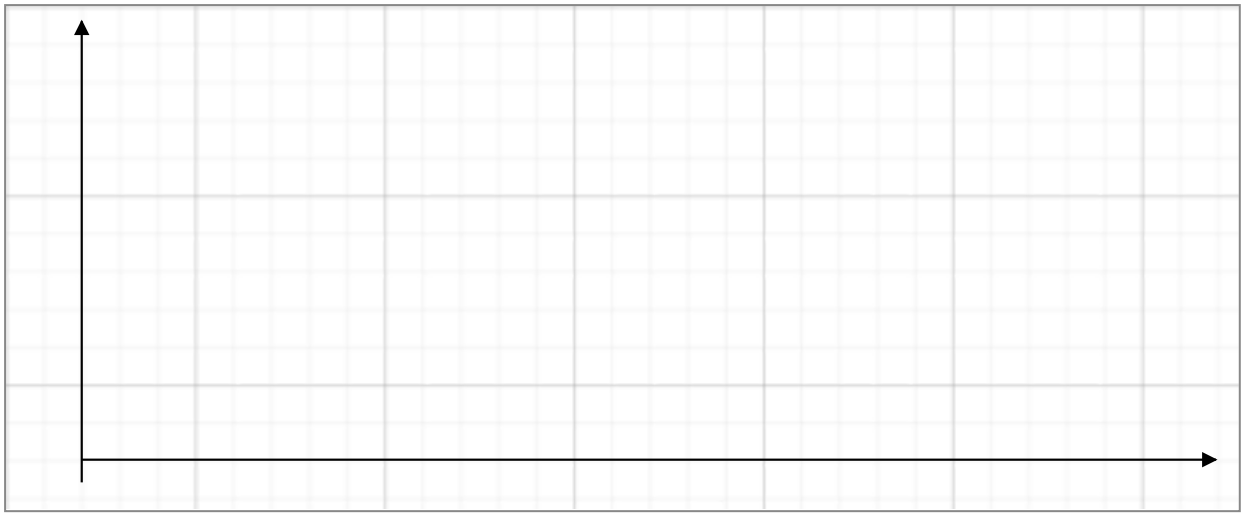
B.4 (0.3pt)

Condition:

B.5 (1.5pt)

Conditions:

Plot:



B.6 (0.5pt)

$I(V_g, V) =$

Part C. Tunneling at finite temperature (2.5 points)

C.1 (0.5pt)

$C_1 = C_2 =$ $C_g =$

C.2 (0.3pt)

$dp(n)/dt =$

C.3 (0.5pt)

$p(n+1) =$

C.4 (0.3pt)

C.5 (0.5pt)

$p(0) =$ $p(1) =$

C.6 (0.4pt)

$I =$